

Introduction

We are looking to develop a Customer Engagement Platform that enables businesses to engage with their customers more effectively through targeted messaging and real-time interactions. The platform should leverage live segmentation and campaign triggers based on specific customer behaviors.

Objective

Your task is to design the architecture and core functionalities of a system that supports these capabilities. You should consider scalability, reliability, and usability in your design. Your design document should outline the approach to data management, system architecture, APIs, and any relevant algorithms or technologies needed.

Key Features

Live Segmentation:

- Dynamically segment users based on real-time data such as location, behavior, device usage, and more
- Ensure the segmentation process is efficient and can handle large volumes of data without significant latency

Behavioral Triggers:

- Implement triggers that initiate campaigns based on specific user actions (e.g., if a user adds an item to a cart but doesn't check out within 1 hour)
- Allow for customization of triggers by end-users of the platform

Campaign Management:

- Design a system to create, manage, and monitor campaigns that send personalized messages based on the defined **Behavioral Triggers**
- Include capabilities for users to define message formats, select communication channels (e.g., SMS, email, push notifications), and set campaign objectives

Analytics and Reporting:

- Provide comprehensive analytics on campaign performance, user engagement, and segmentation effectiveness
- Ensure the platform can generate real-time reports and has the ability for deeper analytical insights with data export functionalities

Deliverables

Technical Requirements Document (TRD):

A detailed document that outlines the system architecture, data model & flow, integration points, and technical stacks proposed.

System Design Diagrams:

High-level architecture diagrams and any detailed component or process diagrams if needed

API Specifications:

Define key API endpoints that will be used within the platform, including methods, request/response formats, and error handling

Mock-ups (optional):

Visual mock-ups or wireframes of the user interface, if time allows

Evaluation Criteria

Completeness of the Design: Coverage of all specified features.

Scalability and Performance: How well the design accommodates growth and performs under load.

Innovation and Practicality: Creativity in the solution and practicality of implementation.

Clarity and Detail: Clarity of communication and level of detail in the documentation.